GIACOMO GUARESI

Full-Stack Developer

Web applications for machines, plants and industrial data

Full-stack developer with an industrial background: eight years of software in production, from GMP pharmaceutical plants to an industrial 3D printer. Today I build and run the internal platform of a manufacturing startup end to end — React, Node.js, PostgreSQL, Docker and CI/CD — and look after the software of its machine. I work best where web applications meet equipment, field protocols and process data. Milan area, Italy — open to hybrid or remote roles.

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SKILLS

Core	React · TypeScript · Node.js · C# / .NET · PostgreSQL · Docker · CI/CD (GitHub Actions, GitLab CI) · Linux servers, production deploys, domains, TLS
Also	Angular · Python · Java / Spring · C / C++ · Kubernetes (as a user of managed clusters) · Kafka · Redis pub/sub · MongoDB · OAuth · Microservices and event-driven services · LLM integration with function calling (Gemini) · Supabase · Query tuning on datasets up to ~1 TB
Industrial & IoT	Modbus · OPC-UA · MQTT · CAN bus · SCADA (iFIX, Movicon) · ESP32 · STM32 · Embedded Linux · PCB design · 3D printing (Klipper, OrcaSlicer)

EXPERIENCE

Ginger Additive — Software Engineer (Full-Stack) `2024 — present`

- Designed, built and maintain, as the only developer, the company's internal platform: around ten web applications in **React**, **Node.js** and **PostgreSQL** (CRM, ERP, project management, production, warehouse, customer support) used every day by all 8 people in the company.
- Defined its architecture — one central database and a shared template every new application starts from — and set priorities together with management. Review and approve about 8 pull requests a week that non-developer colleagues (marketing, production, management) write with AI.
- Own the infrastructure underneath: Linux servers, Docker deployments, CI pipelines, domains and TLS; around ten production releases a month on average.
- Built an AI assistant inside the project management tool: Gemini with function calling creates and updates tasks and answers questions about projects in plain language, within the login and roles of the person using it. In daily use.
- Also responsible for the software of the G1 industrial 3D printer, 100+ machines in the field: control stack on Klipper and custom Linux images, 20+ PCBs designed and taken to production (5 in every G1), and upstream contributions: 8 PRs merged into OrcaSlicer (C++), others into Kalico (the Klipper fork), MainsailOS and KlipperScreen.

Intre — Full Stack Developer (consulting) `2023 — 2024`

- Feature development, front end and back end, on the web product of a large industrial client, in an international agile team of 4 working entirely in English (details confidential).
- **React**, **Angular** and **TypeScript** front ends over containerised services and APIs; deployed and looked after my own services on a **Kubernetes** cluster run by another team.

AM Instruments — Software Developer, industrial monitoring & R&D `2019 — 2023`

- Built the **.NET** web portal and the iOS and Android apps for a room sterilisation system used in pharmaceutical plants, plus its Python tooling; on site in at least 20 pharmaceutical companies across Italy.

- Tuned databases and process historians approaching a terabyte (indexing, query optimisation) for monitoring systems with hundreds of sensors: yearly data extraction cut from a day of processing to about an hour.
- Integrated monitoring software (SCADA: iFIX, Movicon) with plant equipment over Modbus, OPC-UA, Profinet and EtherCAT, under GMP and ISO 9001.
- Predictive maintenance and adaptive thresholds on process data, in production in the system shipped to customers.
- Wrote the embedded C++ for the sterilisation controller (ARM, embedded Linux, CAN bus).

Interlem — Full Stack Developer 2018 — 2019

- Built a **Spring** and **PostgreSQL** web portal for managing and reordering a fleet of IoT-connected gas cylinders (LoRa smart scales, hardware built in a team).
- Refactored a legacy Delphi business portal and maintained it until its .NET replacement shipped.

PROJECTS

Open source	8 PRs merged into OrcaSlicer (C++); one each into Kalico, MainsailOS and KlipperScreen.
FAInder	Mobile app (React Native / Expo, TypeScript) for exploring Italy's FAI heritage sites on a map, with check-ins and a nightly data refresh on GitHub Actions. github.com/GiacomoGuaresi/FAInder
Personal PWAs	Two PWAs in React, TypeScript and Supabase, real-time and offline, with Vitest suites, written architecture and decision records, and automated deploys. Grocery · Projects

EDUCATION & LANGUAGES

Diploma	Perito Informatico (technical diploma in computer science) — ITIS Hensemberger, Monza, 2018
Languages	Italian (native) · English (C1, daily working language in international teams)
Learning	Rust · Agentic AI systems